ENGINEERING PORTFOLIO

Fatma Nilgun Abali

Mechanical engineering graduate, UBC. Mechanism design, prototyping, and testing.

01 6-DoF Motion Platform

02 Dometic Marine — Product Design Co-op

03 UBC Agrobots — Weed Sprayer

04 Other Work



6-DoF motion platform: the 1:4 prototype, built and tested

fatmanilgunabali@gmail.com

linkedin.com/in/fatma-nilgun-abali

github.com/NilgunAbali

6-DoF Motion Platform — Overview and results

Servotest builds lab rigs that reproduce road, track and earthquake motion. Our brief: a lightweight 6-DoF platform for race-car suspension testing, sized at full scale for a 500 kg payload at up to 200 Hz. In a conventional Stewart platform the actuators are the legs, so their mass moves with the load and limits speed. We grounded all six and drove the platform through bell cranks and pushrods.

TESTED ON THE 1:4 PROTOTYPE

14% of total mass is moving mass (target under 15%)

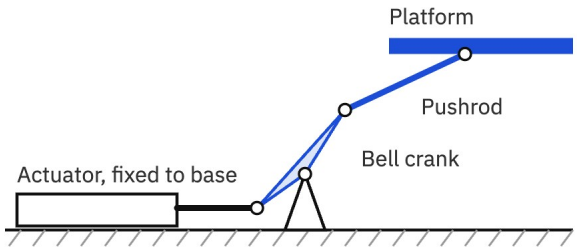
5.6 kg payload carried in all directions (target 5 kg)

IMU measured rotations matched the commanded poses

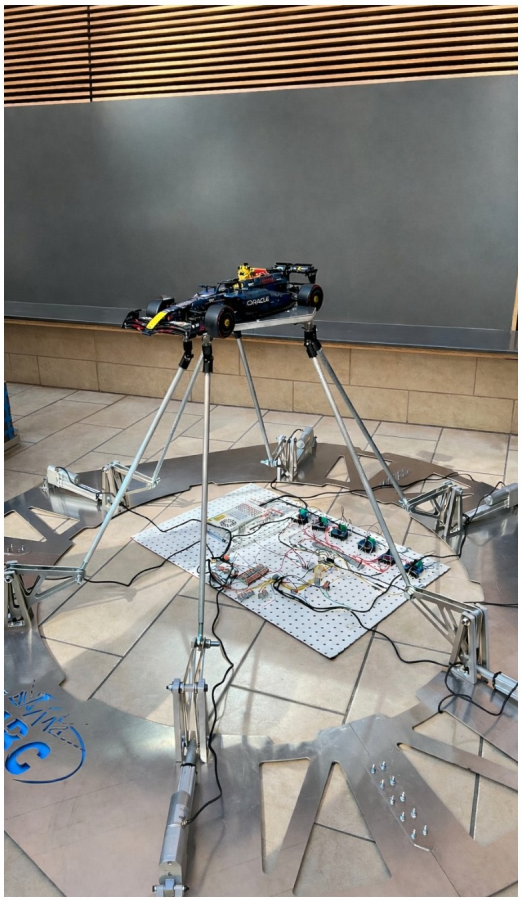
MY ROLE

- Sourced materials and fabricated the prototype: waterjet, 3D printing, machining
- Sized and selected the actuators
- Built and wired the control board
- Integrated an IMU for validation testing
- Produced the engineering drawings, schematics, and project poster

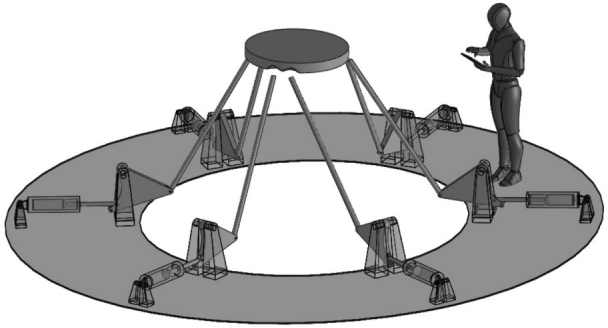
HOW IT WORKS



One of six legs. Only the blue parts move.



1:4 prototype, built and tested



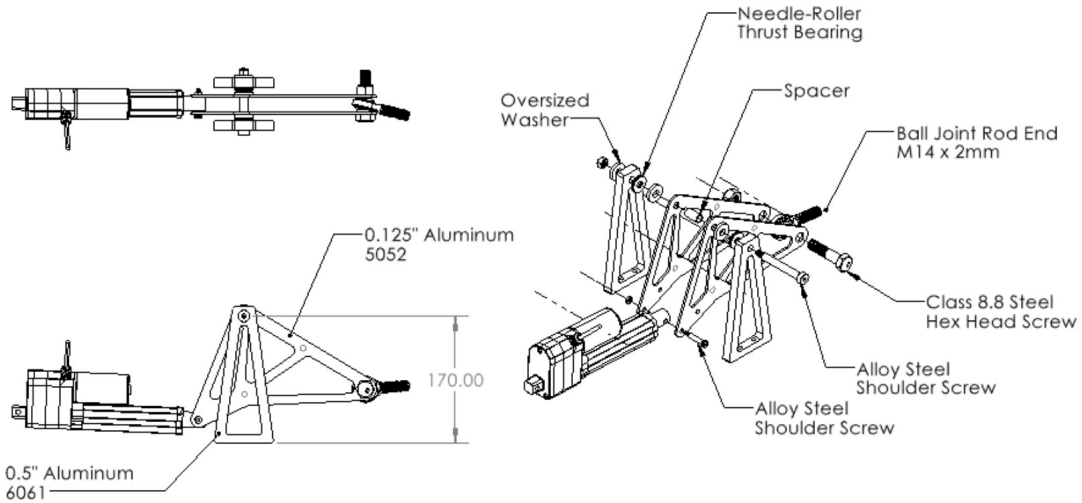
Full scale, virtual model: 500 kg, 200 Hz



1:4 prototype, CAD. Same human figure in both renders.

6-DoF Motion Platform — Mechanism design and drawings

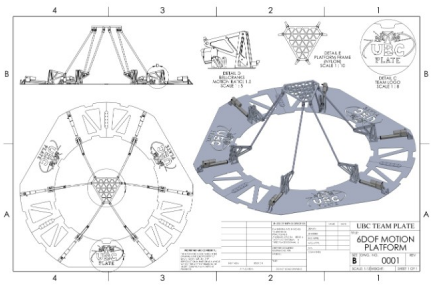
Every part detailed in SolidWorks for waterjet cutting and machining, from the full assembly down to each sub-assembly.



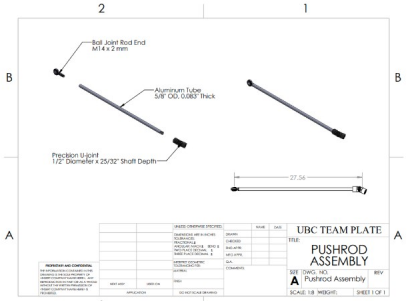
Bell-crank assembly: materials and hardware called out

Sizing the actuators

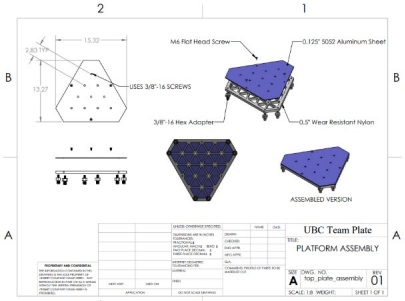
Duty cycle	Highest available
Resolution	Highest available
Stroke	Matched to the platform's range of motion
Limits	Cost and supplier lead time
Choice	After team-wide research, we chose the 12 V actuator I found, from a Vancouver supplier



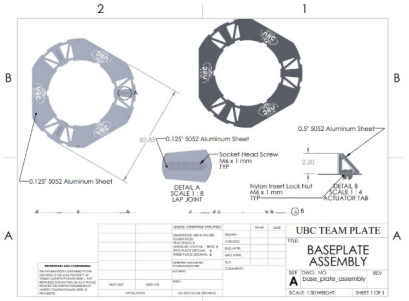
Full assembly, 1:15



Pushrod



Platform



Base plate

6-DoF Motion Platform — Fabrication and wiring

Waterjet-cut aluminium base and bell cranks; fittings milled, turned, tapped and die-threaded.



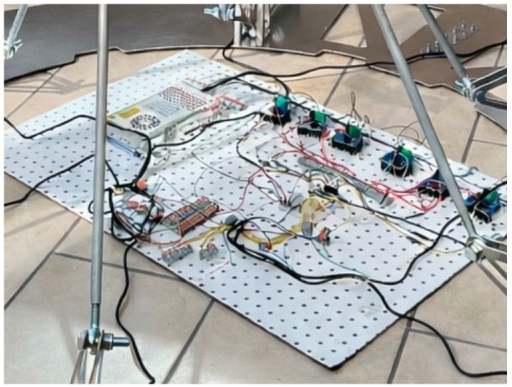
Pushrod, 2024
aluminium



Actuator bolted to the base, so its mass never moves with the platform

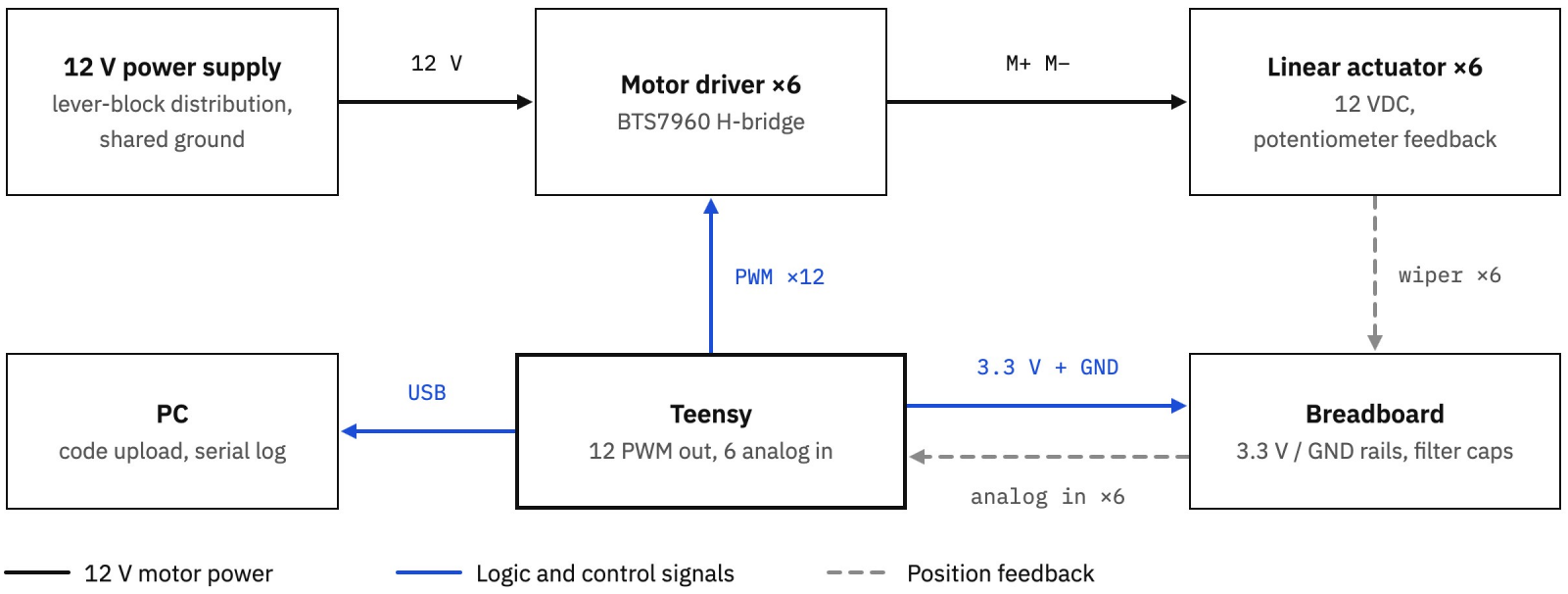


Aluminium top plate on a nylon frame, keeping the moving mass low



Teensy and six motor drivers, wired with a teammate

CONTROL BOARD WIRING

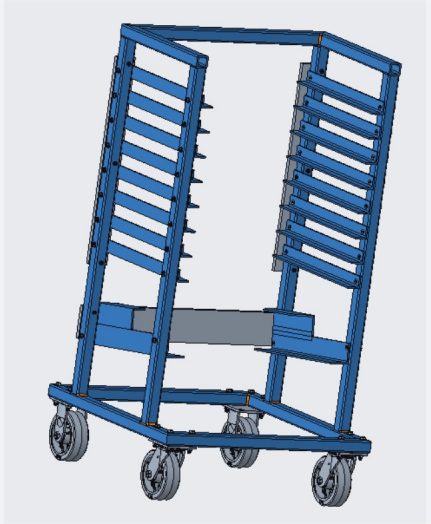


02 · PRODUCT DESIGNER, MECHANICAL CO-OP · FEB – AUG 2025

Dometic Marine — Shop equipment design and load checks

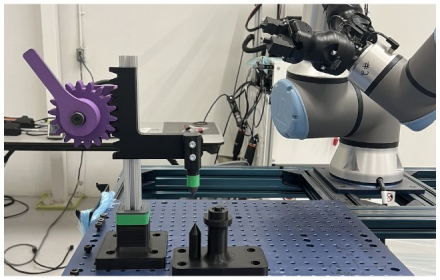
Dometic's marine division designs and builds SeaStar steering systems for boats. Over two co-op terms in the machining department I designed shop equipment, released drawings and checked loads.

Ergonomic finished-goods cart: $10K redesign proposal



Creo model of the redesigned cart

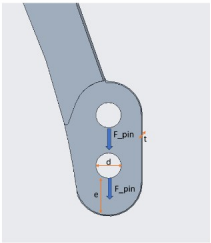
Problem	Operators found it hard to push; the safety officer made it my project
Standard	WorkSafeBC limits on pushing loads and time; met with upgraded casters and fewer tray slots
Design	Reverse-engineered the old cart; 198-part welded Creo assembly; 4-sheet GD&T drawing set
Quotes	From machine shops across Metro Vancouver
Proposal	$10K, buying the carts over 2–3 years
Outcome	Presented to the machining manager, who took it to the safety committee; handed it off to supervisor for procurement



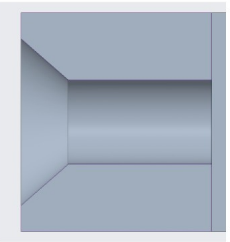
Robotic inspection feasibility

Designed and 3D-printed a motion-replication fixture and programmed a UR arm to test whether CMM inspection could be automated. It was feasible at this stage; the next step is prototyping how the UR integrates into the inspection cell.

Lifting-cradle stress checks



Clevis



Screw section

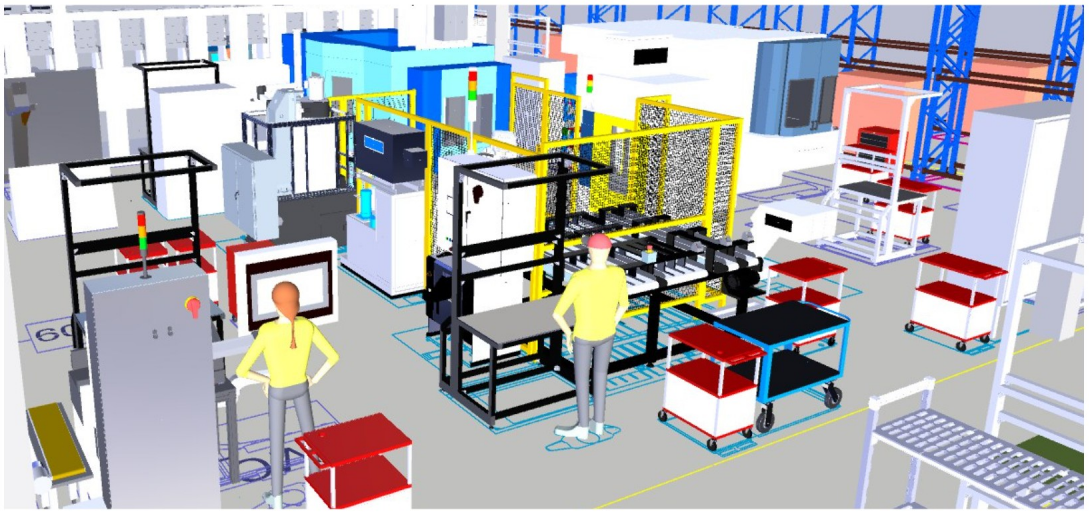
Check	Stress, psi	SF
Clevis pin tear-out τ = F / 2t(e – d), von Mises	1,397	28.6
Clevis bearing σ = F / dt, vs 27 ksi allowable	538	50.2
Frame, shear + bending	4,928	8.1
Screw section, tension + eccentric bending	8,202	**4.9**

Contributed to the stress checks for a cradle lifting a 400 lb flywheel, in Excel. Al 6061-T6, 40 ksi yield; F = 100 lbf per clevis pin.

02 · PRODUCT DESIGNER, MECHANICAL CO-OP · FEB – AUG 2025

Dometic Marine — Production floor improvements

Working alongside operators to make 50 CNC machines easier to monitor, move and maintain.



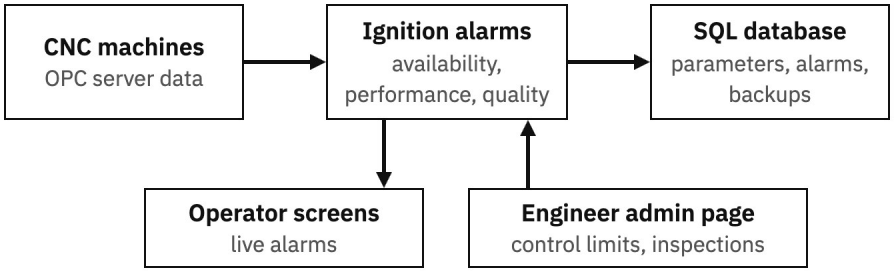
Factory re-layout

- Planned CNC machine moves between buildings from field measurements of machines and surrounding equipment
- Chose new machine locations based on space, access and workflow
- Kept the plant's 2D AutoCAD layouts and 3D Navisworks model up to date after each move

Ignition SCADA for 50 CNC machines

- Programmed front- and back-end alarm logic on OPC server data: availability, performance (under 85% of expected rate), quality (over 5% and 10% fail rate) and checklist alarms
- Stored machine parameters and alarm records in SQL, and set up database backups
- Built an engineer-facing admin page for editing control limits and in-cell inspections
- Refined alarm thresholds and screens with operator feedback
- Refactored a 2,000-line script for cleaner SCADA integration

SCADA DATA FLOW



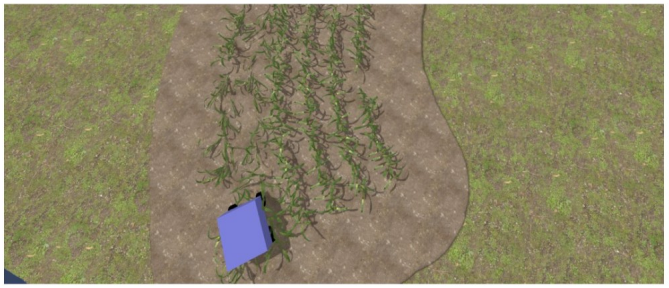
Specialty wrench and filter access

Developed a custom filter-access solution and specialty wrench that improved ergonomics and cut weekly maintenance time for operators.

03 · WEED SPRAYING TEAM LEAD JAN 2024 – APR 2025 · TEAM MEMBER SEP 2023 – JAN 2024

UBC Agrobots — Targeted weed-spraying system

UBC Agrobots (ubcagrobot.com) builds autonomous robots for precision agriculture. Its spraying system targets only the weeds its AI detects, which the team estimates cuts herbicide use by up to 80%. I joined in September 2023 and led the 10-student spraying team from January 2024 to April 2025, taking it from concept through to prototyping; the system was built after my term ended.



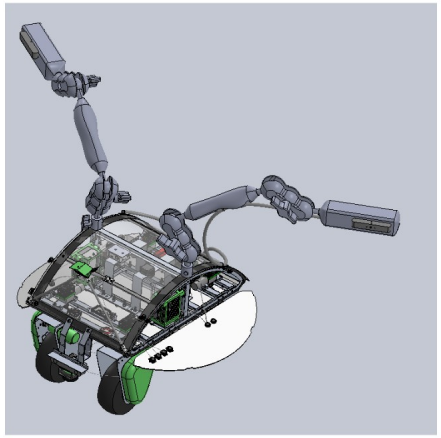
Field simulation concept. Image: UBC Agrobots

HOW TARGETED SPRAYING WORKS · BLUE = MY WORK

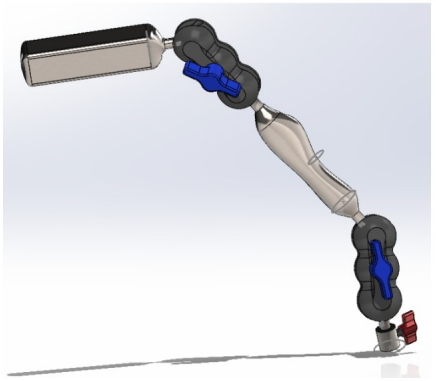
01 · SENSE	02 · DETECT	03 · CIRCUIT	04 · SPRAY	05 · RESULT
ZED stereo cameras aimed by the adjustable arms I designed	**AI weed detection** Applied AI team	**Spray circuit** power budget, safety parts, EasyEDA schematic	**Pump and valve** tested for flow rate, spray range and valve recoil	**Weeds only** up to 80% less herbicide (team estimate)

MY ROLE

- Led a 10-student team from concept generation through risk analysis to prototyping
- Ran a DFMEA, rating failure modes by severity, occurrence and detectability to prioritize fixes
- Designed the spray circuit: power budget, safety components and EasyEDA schematic
- Wrote and ran a 10-step test plan on the pump and circuit; the flow rate, spray range and valve recoil results sized the circuit equipment
- Designed a 3D-printable camera support with 6-DoF adjustability in SolidWorks, so the ZED cameras could be aimed for weed detection
- Coordinated interfaces with the Chassis, Applied AI and Navigation teams



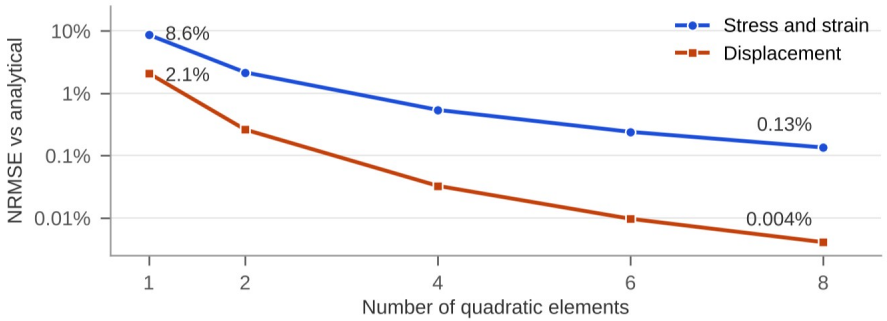
Adjustable arms holding the ZED cameras



Ball joints with locking knobs aim each camera in 6 DoF

Solvers and Abaqus files for both FEM projects: github.com/NilgunAbali

Other Work — Structural analysis and earlier projects

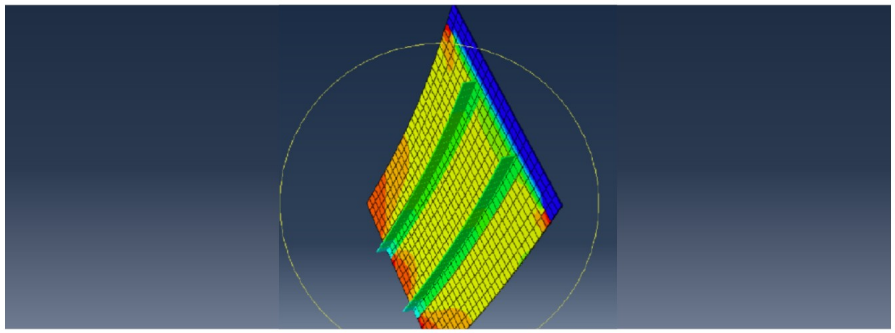


My Python solver: error vs the analytical solution

TU Delft exchange · Linear Modelling · Sep – Oct 2024

Ingenuity Mars helicopter: FEM and convergence

- Derived analytical and FEM models of the blade under linearly varying centrifugal load, using quadratic bar elements
- Wrote a Python FEM solver with a variable element count, and reran it in a convergence study to find the optimum mesh
- Simulated the main beam under worst-case loading in Abaqus



My Abaqus model: wing skin with two L-stiffeners

TU Delft exchange · Abaqus project · Nov 2024

Airplane wing under lift: stiffener study

- Modelled a wing under lift in Abaqus, using the wing's symmetry to simplify the model
- Compared structural modifications to find the layout with the least displacement
- Compared two ways of attaching the L-shaped stiffeners to the skin: tie constraints and merged geometry

UBC Solar · Battery Mechanical · Sep 2021 – Dec 2022

Battery-removal jig for a solar race car

Designed a SolidWorks jig for removing the battery from the race car, applying DFM/DFA to cut assembly time and improve safety. Also instrumented, spot-welded and soldered sensors, LEDs and circuitry onto PCBs.

UBC Pulp and Paper Centre · Research Assistant · May – Aug 2023

Pilot-scale pulp screening research

Analyzed LabVIEW sensor data in MATLAB and Excel, computing 100+ stress and strain parameters to find optimal screening conditions. Traced a critical screen failure to its root cause, saving two weeks, and trained 6 researchers.